

Salvatore Danilo Palumbo

 GitHub |  LinkedIn |  Portfolio
 Mail |  Favara, Sicily, Italy
Preferred name: **Danilo**

SUMMARY

Data Scientist and ML / AI Engineer with a strong research background and experience in AI since 2017. Skilled in designing, developing, and deploying production-ready Machine Learning systems across NLP, geospatial data, Generative AI, and AI agents.

PROFESSIONAL EXPERIENCE

Founding Data Scientist / ML Engineer / Researcher

Nov 2024 - Jul 2026

IdentifAI Labs

Joined the startup as first Data Scientist / ML Engineer, contributing to the foundation and growth of the ML models and functions. Responsible for developing and researching advanced ML for Deepfake voice detection, leveraging complex datasets and diverse audio formats. Contributed to the evolution of the production platform through feature design, automated testing, and robust use-case development. Drove cutting-edge R&D in audio signal processing and computer vision. Oversaw end-to-end research initiatives, managing the creation and deployment of ML models for AI and Deepfake detection. Building models and techniques for tampering detection in documents.

Data Scientist

Mar 2022 - Nov 2024

TopNetwork S.P.A.

Handled the full lifecycle of Open Data, Italian government datasets, and geospatial data, from cleaning and wrangling to analysis and enrichment. Developed interactive dashboards using Dash-Plotly and implemented high-quality APIs with FastAPI. Managed databases and object storage systems, and performed satellite imagery processing, including the computation of indices such as NDVI and NBR. Designed and maintained ETL pipelines for large-scale geospatial workflows.

Freelancer Data Scientist / Data Engineer / ML Engineer

Jan 2020 - Feb 2022

Delivered custom machine learning models using Scikit-learn and TensorFlow, along with targeted data analysis and visualization solutions. Executed advanced web scraping and built automation tools and bots across multiple platforms. Developed ETL pipelines in Python with SQLAlchemy and provided consulting on software development and Python best practices. Offered tutoring in Mathematics, Computer Science, and English.

Curricular Internship – Haden Digital di Carmelo Pullara

2020 - 2021

Basics of containers with Docker and Kubernetes. Basics of Game Development with Unity and C#.

Curricular Internship – Haden Digital di Carmelo Pullara

2018 - 2019

Basics of Web Development with HTML, CSS and Javascript. Basics of Web frameworks and other tools like Bootstrap, Wordpress, SASS/SCSS etc.

EDUCATION

2019 - 2021 Master's Degree in Computer Science at **Università Degli Studi Di Palermo**

2016 - 2019 Bachelor Degree in Computer Science at **Università Degli Studi Di Palermo**

2010 - 2015 High School Diploma at **I.T.C.E.T. "Leonardo Sciascia"**

ACADEMIC ACHIEVEMENTS AND PUBLICATIONS

Rocchesso, D., Andolina, S., Ilardo, G., Palumbo, S. D., Galluzzo, Y., & Randazzo, M. (2022). A perceptual sound space for auditory displays based on sung-vowel synthesis. *Scientific Reports*, 12(1), 1-13.

Attestation of Proficiency in Data Science

Bachelor thesis in collaboration with the company "Open Content"

TECHNICAL SKILLS

Programming Languages	Python, SQL
Machine Learning & AI	PyTorch, PyTorch Lightning, TensorFlow, Scikit-learn, HuggingFace, Pandas, NumPy, SciPy
LLM & NLP Frameworks	LangChain, LlamaIndex, Qdrant, ChromaDB, FAISS, LiveKit
Data & Experimentation Tools	MLOps, Web Scraping, Data Analysis, Machine Learning Pipelines
Web Development (Basic)	HTML, CSS, JavaScript, Bootstrap, WordPress
Additional Languages (Familiarity)	C, Java, Dart (Flutter)
Research Interests	Artificial Intelligence, Machine Learning, Natural Language Processing, Large Language Models, Computer Vision, Deepfake Detection, Satellite Data Analysis

OTHER INTERESTS

Gaming, Origami, Technology, Chess, Music, UI/UX